



United International University
School of Science and Engineering
Department of Computer Science and Engineering
Final Examination Trimester- Summer – 2024
Course: Math -2205
Total marks - 50; Duration – 2 hours

[Note that the number of marks is given in brackets [] at the end of each question or part question. You are requested to answer all the questions in order.]

Q1

- (a) Two ordinary fair dice are rolled. Using a possibility diagram or otherwise, find the probability of obtaining
- at least one 6
 - two numbers whose product is 6.
- (b) A coin is biased such that the probability that three successive tosses all result in heads is $\frac{125}{512}$. Find the probability of tail to appear when tossed.
- (c) In a survey, 50% of the participants own a desktop (D), 60% own a laptop (L) and 15% own both. By using Venn diagram or otherwise, find what percentage of participants owns neither a desktop nor a laptop?
- (d) The following table gives information about all the animals on a farm. Find the probability that a randomly selected animal is:

	Male	Female
Goats	5	25
Sheep	3	22

- male or a goat
- a sheep or female

[4+2+2+4 = 12]

Q2

- (a) A factory produces half-liters tins of oil. The volume of oil in a tin is normally distributed with a mean 506 ml and standard deviation 2.9 ml. A tin is randomly selected what is the probability that it will be
- less than half a liter (500 ml) of oil.
 - Within one standard deviation
- (b) A footballer has a 95% chance of scoring each penalty kick that she takes. Find the probability that she:
- scores from all of her next 10 penalty kicks
 - fails to score exactly 2 of her next 10 penalty kicks.

- (c) The probability distribution for random variables Y is given in the following table

y	0	1	2	3	4
$P(Y=y)$	0.03	$2p$	0.32	p	0.05

- Find the value of p .
- Find the standard deviation of Y

[6+4+4 = 14]

Q3

- (a) The weekly petrol consumption, in hundreds of liters of a sales representative, may be modeled by the random variable X with probability density function (pdf)

$$f(x) = \{a x(4 - 3x)\} \text{ for } 0 \leq x \leq \frac{4}{3}.$$

- Find the value of a .
- Find the mean of the weekly consumption
- Find the probability that the weekly consumption will be less than 1 hundred liters.
- Find the mode of the weekly consumption.
- Construct the corresponding cdf for the above pdf

[2+3+3+3+3 = 14]

Q4

- (a) A machine is supposed to produce metal rods which are 5.7 cm long. A random sample of 100 rods produced by the machine are measured with mean 5.71 cm and standard deviation 0.048 cm. Calculate a 95% confidence interval for the mean length of a rod produced by the machine.
- (b) A logistics company claims that the average delivery time for packages is 5 days. A customer service team wants to test if the actual average delivery time is longer than this claim. The delivery times follow normal distribution with a known standard deviation of $\sigma = 1.5$ days. A random sample of 40 deliveries shows an average delivery time of 5.4 days. Perform a one-tailed test at a 5% level of significance to determine if the true average delivery time exceeds 5 days, [use σ instead of $\sigma/\sqrt{n}$.]
- (c) State possibly what kind of error may occur in this situation.

[3+6+1 = 10]

Important Formulae

Distribution

pmf/pdf

Binomial

$$f(x) = nC_x p^x (1-p)^{n-x}; x = 0, 1, 2, \dots, n$$

Poisson

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}; x = 0, 1, 2, \dots$$

Uniform

$$f(x) = \frac{1}{b-a}; a \leq x \leq b$$

Normal

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty < x < \infty$$

For continuous random variables

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$\text{Var}(x) = \int_{-\infty}^{\infty} x^2 f(x) dx - (E(x))^2$$

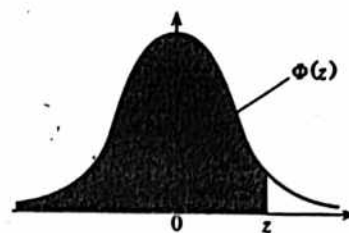
$$CI = \bar{x} \pm z \frac{s}{\sqrt{n}}$$

The normal distribution function

If Z has a normal distribution with mean 0 and variance 1 then, for each value of z , the table gives the value of $\Phi(z)$, where

$$\Phi(z) = P(Z \leq z).$$

For negative values of z use $\Phi(-z) = 1 - \Phi(z)$.



z	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
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0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359	4	8	12	16	20	24	28	32	36
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753	4	8	12	16	20	24	28	32	36
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141	4	8	12	15	19	23	27	31	35
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517	4	7	11	14	18	22	25	29	32
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879	4	7	11	14	18	22	25	29	32
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224	3	7	10	14	17	20	24	27	31
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549	3	7	10	13	16	19	23	26	29
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852	3	6	9	12	15	18	21	24	27
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133	3	5	8	11	14	16	19	22	25
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389	3	5	8	10	13	15	18	20	23
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621	2	5	7	9	12	14	16	19	21
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830	2	4	6	8	10	12	14	16	18
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015	2	4	6	7	9	11	13	15	17
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177	2	3	5	6	8	10	11	13	14
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319	1	3	4	6	7	8	10	11	13
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441	1	2	4	5	6	7	8	10	11
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545	1	2	3	4	5	6	7	8	9
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633	1	2	3	4	4	5	6	7	8
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706	1	1	2	3	4	4	5	6	6
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767	1	1	2	2	3	4	4	5	5
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817	0	1	1	2	2	3	3	4	4
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857	0	1	1	2	2	2	3	3	4
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890	0	1	1	1	2	2	2	3	3
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916	0	1	1	1	1	2	2	2	2
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936	0	0	1	1	1	1	1	2	2
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952	0	0	0	1	1	1	1	1	1
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964	0	0	0	0	1	1	1	1	1
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974	0	0	0	0	0	1	1	1	1
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981	0	0	0	0	0	0	0	1	1
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986	0	0	0	0	0	0	0	0	0